

An evaluation of ECMWF SEAS5 seasonal climate forecasts for Australia using a new forecast calibration algorithm

Q.J. Wang^{a,*}, Yawen Shao^a, Yong Song^b, Andrew Schepen^c, David E. Robertson^b, Dongryeol Ryu^a, Florian Pappenberger^d

^a Department of Infrastructure Engineering, The University of Melbourne, Parkville, 3010, Australia

^b CSIRO Land and Water, Clayton, 3168, Australia

^c CSIRO Land and Water, Dutton Park, 4102, Australia

^d European Centre for Medium-range Weather Forecasts (ECMWF), Reading, RG2 9AX, UK

ARTICLE INFO

Keywords:

Forecast calibration and evaluation
Forecast skill
Forecast reliability
Precipitation
Temperature

ABSTRACT

The commencement of SEAS5 model for operational seasonal climate forecasting by the European Centre for Medium-Range Weather Forecasts (ECMWF) is a new development. It replaces the ECMWF System 4, which had a large international community of seasonal climate forecast users. To assist potential users of SEAS5 forecasts, a systematic and detailed evaluation of forecast skill and reliability of climate variables over land areas is valuable. In this regional study, we evaluate SEAS5 performance in forecasting precipitation and daily minimum temperature (Tmin) and daily maximum temperature (Tmax) for the Australian continent, based on 36 years of re-forecast data. We evaluate forecasts after simple mean-corrections, and statistically calibrated forecasts using the Bayesian joint probability (BJP) modelling approach. We also provide a comparison with System 4. A new simpler and more efficient BJP algorithm is introduced to facilitate this study and support wider use of the algorithm in other applications.

1. Introduction

Australia has a highly variable climate. Skillful, reliable and high-resolution seasonal climate forecasts from global circulation models (GCMs) are in demand for supporting proactive decision-making in many climate-sensitive sectors (Troccoli, 2010). The Australian Bureau of Meteorology (BoM) recently upgraded its seasonal climate forecasting model to ACCESS-S1, and is beginning to develop its next version, ACCESS-S2 (Hudson et al., 2017). A limitation of ACCESS-S1 is its relatively short re-forecast period (from 1990). It is planned that ACCESS-S2 will have a longer re-forecast period (from 1981).

Internationally, a significant recent development in seasonal climate forecasting is the commencement of SEAS5 from the European Centre for Medium-Range Weather Forecasts (ECMWF). SEAS5 re-forecasts are available for 1981 onwards. Compared to its predecessor System 4, SEAS5 is a substantially changed forecast system. It includes upgraded versions of the atmosphere and ocean models at higher resolutions, and adds a prognostic sea ice model. Johnson et al. (2018) reported major improvements in performance from System 4 to SEAS5, based on some

large-scale climate and ocean diagnostics. Emerton et al. (2018) demonstrated the value of SEAS5 in global hydrological seasonal forecasting.

ECMWF System 4 became operational in 2011 (Molteni et al., 2011). Its forecasting performance has been extensively evaluated. Kim et al. (2012) assessed temperature and precipitation forecasts for Northern Hemisphere winter at the global scale. Forecasts were found to be the most skillful in the tropics, especially in the El Nino region. Weisheimer and Palmer (2014) evaluated the reliability of temperature and precipitation forecasts worldwide. They assigned ratings from 5 (perfect) to 1 (dangerous) to the forecasts for different parts of the world. They found that for surface temperature, and even more for precipitation, forecast probabilities are not reliable when different from climatology and away from the El Nino region.

The commencement of ECMWF SEAS5 is a new development. To assist potential users of SEAS5 forecasts, a systematic and detailed evaluation of forecast skill and reliability of climate variables over land areas is valuable. In this regional study, we evaluate SEAS5 performance in forecasting precipitation, daily minimum temperature (Tmin) and

* Corresponding author.

E-mail address: quan.wang@unimelb.edu.au (Q.J. Wang).

<https://doi.org/10.1016/j.envsoft.2019.104550>

Received 8 March 2019; Received in revised form 22 August 2019; Accepted 9 October 2019

Available online 18 October 2019

1364-8152/© 2019 Elsevier Ltd. All rights reserved.

daily maximum temperature (Tmax) for the Australian continent, based on 36 years of re-forecast data. We compare its performance with System 4.

There are significant benefits from post-processing climate forecasts. Raw GCM forecasts are biased, unreliable in ensemble uncertainty spread (too wide or too narrow), and often less skillful than the naïve climatology forecasts (Shukla and Lettenmaier, 2013; Schepen and Wang, 2014). Post-processing aims to resolve these problems, while extracting the most out of the underlying forecast skill of the GCMs (Zhao et al., 2017). Examples of statistical post-processing methods are Gneiting et al. (2005), Wilks and Hamill (2007), Unger et al. (2009), Schepen et al. (2014), Zhang et al. (2017) and Narapusetty et al. (2018). In this study, we apply the Bayesian joint probability (BJP) model (Wang et al., 2009; Wang and Robertson, 2011) to calibrate the SEAS5 forecasts, and evaluate SEAS5 performance both before and after the calibration.

The BJP model has been extensively applied to calibrate seasonal forecasts (Hawthorne et al., 2013; Peng et al., 2014; Schepen and Wang, 2014; Schepen et al., 2014, 2016, 2018; Bennett et al., 2016, 2017; Zhao et al., 2017, 2019; Strazzo et al., 2018). In the BJP model, a joint probability distribution is used to characterize the relationship between the raw GCM ensemble means and observations. The distribution is subsequently used to produce calibrated forecasts by conditioning the distribution on new raw forecasts. While the original BJP model was developed to deal with a number of data issues, it is used here primarily for its ability to deal with zero values that can occur with both raw forecasts and observations of precipitation, either separately or concurrently (see Section 3.1 later).

While the BJP method has worked well in the past, the original mathematical formulation is onerous, and computer coding for its numerical implementation is demanding. In this paper, we introduce a new BJP algorithm that is much simplified in its mathematical formulation and more straightforward to code. Importantly, it is computationally much more efficient. For large-scale applications such as over the Australian continent, calibration involves a huge number of grid-cells and many lead-times, and repeated applications are required for rigorous cross-validation. The computation effort is large, especially for high-resolution models such as the SEAS5. The development of the new BJP algorithm makes post-processing easier and faster.

2. Data

2.1. GCM forecasts

ECMWF System 4 and SEAS5 ensemble forecast datasets for the Australian continent are used in this study. System 4 was introduced in November 2011, benefiting from a NEMO ocean model and an ocean data assimilation system (NEMOVAR). It was a fully coupled ocean-atmosphere dynamical forecasting system. The horizontal resolution of the Integrated Forecast System (IFS) atmosphere module was approximately 80 km. The ERA-Interim reanalysis data provided the ocean module with forcing fluxes and the atmosphere module with unperturbed initial conditions. Ocean initial conditions were derived from ORA-S4 for the re-forecasts and from the real-time NEMOVAR for the operational period. Perturbed initial conditions and stochastic physics were used to generate ensembles for the re-forecasts and operational forecasts.

SEAS5 possesses improved atmosphere and ocean model configurations with higher resolutions, approximately 36 km in IFS horizontal resolution, and includes an interactive sea ice model, LIM2. The atmosphere initialization of SEAS5 is the same as System 4, while its ocean and sea-ice initial conditions are supported by a new operational ocean analysis system, OCEAN5 (Zuo et al., 2018). This ocean system is composed of ORAS5 for historical reanalysis and OCEAN5-RT for daily real-time analysis. Unperturbed and perturbed atmospheric initial conditions, and stochastic model perturbations are applied to generate

ensemble members.

The re-forecasts and operational forecasts for System 4 were initialized on the 1st of every month and run for 7 months. The re-forecast period was 1981–2010, with an ensemble size of 15 members. The real-time forecast period was 2011–2017, with an ensemble size of 51 members. SEAS5 has the same initialization date and forecast horizon as System 4. Its re-forecast period is 1981–2016, with an ensemble size of 25 members. The real-time forecasts started in 2017, with an ensemble size of 51 members.

In this study, we evaluate SEAS5 based on the re-forecasts for 1/1/1981–1/12/2016 using 25 ensemble members. For comparison, we use a combination of re-forecasts and operational forecasts of System 4 for the same period. Only the first 15 ensemble members of the operational forecasts of System 4 are used, for consistency with the re-forecasts of System 4.

Target variables of forecasts in this study are monthly and seasonal totals of precipitation, and monthly and seasonal averages of Tmin and averages of Tmax. To compare the performance of SEAS5 with System 4, the SEAS5 data (resolution) are re-gridded to the coarser grid of System 4 (resolution).

2.2. Observed data

Monthly and seasonal observation data are obtained from the AWAP (Australian Water Availability Project) climate datasets of the Bureau of Meteorology (Jones et al., 2009). The AWAP analysis datasets provide daily values of climate variables at high spatial resolution ($0.05^\circ \times 0.05^\circ$) for Australia based on interpolation of gauge data. The data period used in this study is 1/1/1981–30/06/2017. The AWAP data are re-gridded to the grid of System 4.

3. Methods

3.1. The Bayesian joint probability (BJP) model

The ensemble mean of a raw GCM forecast for a climate variable is denoted by y_1 , and the corresponding observation by y_2 . In mathematical terms, we call y_1 a predictor and y_2 a predictand. We first apply transformations to normalize the variables to

$$z_1 = \psi(y_1 | \Delta) \quad (1)$$

$$z_2 = \psi(y_2 | \Delta) \quad (2)$$

where ψ denotes a generic normalizing transformation function, and Δ is a vector of transformation parameters. In this study, we apply the log-sinh transformation to precipitation variables and the Yeo-Johnson transformation to temperature variables (Yeo and Johnson, 2000; Wang et al., 2012). We estimate a single “best” set of transformation parameter values for each variable by using a Bayesian maximum a posteriori (MAP) solution (Schepen et al., 2016).

Next, we assume the transformed predictor and predictand to follow a bivariate normal joint distribution

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (3)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the mean vector and covariance matrix respectively. We denote the parameter set as $\boldsymbol{\theta} = \{\boldsymbol{\mu}, \boldsymbol{\Sigma}\}$.

The bivariate normal distribution applies to continuous variables, but in case of precipitation, variables z_1 and z_2 can be subject to a lower threshold of z_1^* and z_2^* corresponding to zero forecast precipitation and zero observed precipitation. To overcome this problem, we treat a threshold data value as a censored data record, with an unknown exact value that is equal to or below the threshold (Wang and Robertson, 2011).

Given a data series, $\mathbf{D} = \{\mathbf{z}(t), t = 1, 2, \dots, n\}$, we use Bayesian infer-

ence to find the posterior distribution of the model parameters

$$p(\theta|\mathbf{D}) \propto p(\theta)p(\mathbf{D}|\theta) \quad (4)$$

where $p(\theta)$ is the prior distribution of the parameters, and $p(\mathbf{D}|\theta)$ is the likelihood. By setting an appropriate prior distribution, it is possible to formulate Gibbs sampling to numerically sample $p(\theta|\mathbf{D})$.

Once representative parameter samples are available, the BJP model can be used in predictive mode to calibrate a new forecast $y_1(t^*)$, or $z_1(t^*)$ after transformation. The posterior predictive distribution of $z_2(t^*)$ is given by

$$f(z_2(t^*)) = \int p(z_2(t^*)|z_1(t^*), \theta)p(\theta|\mathbf{D}) d\theta \quad (5)$$

Again, Gibbs sampling can be formulated to numerically sample $f(z_2(t^*))$. By back-transforming each of the sample members, we produce a calibrated ensemble forecast of $y_2(t^*)$.

The derivation and implementation of the Gibbs sampling algorithm, including pseudo code, are given in [Appendix A](#). Compared with the original algorithm of the BJP model ([Wang et al., 2009](#); [Wang and Robertson, 2011](#)), the new algorithm is greatly simplified mathematically and more straightforward to code. Importantly, it is computationally much more efficient.

On rare occasions, we find the calibrated forecasts for precipitation can be unrealistically large. The problem is caused by extremely large raw forecast values from the GCM models, so large that the transformed forecasts are considered improbable according to the marginal distribution of the raw forecasts. We use a pragmatic approach here to circumvent this problem.

Given the marginal distribution, $N(z_1|\mu_1, \sigma_1^2)$, for z_1 , we check if the following is true

$$\Phi(z_1(t^*)|\mu_1, \sigma_1^2) > P_{\text{extreme}} \quad (6)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the marginal distribution, and P_{extreme} is a high non-exceedance probability to indicate an extreme threshold. If true, we set $z_1(t^*)$ back to the extreme threshold

$$z_1(t^*) = \Phi^{-1}(P_{\text{extreme}}|\mu_1, \sigma_1^2) \quad (7)$$

where $\Phi^{-1}(\cdot)$ is the inverse cumulative distribution function of the marginal distribution. We then derive the calibrated forecast for $y_2(t^*)$ by conditioning $z_2(t^*)$ on this adjusted $z_1(t^*)$. In this study, we set $P_{\text{extreme}} = 0.999$.

3.2. Forecast evaluation

We evaluate forecasts for the following target variables

- (1) Monthly total of precipitation, monthly average of Tmin, and monthly average of Tmax, at zero lead-time (i.e. forecast at the start of a month for that month)
- (2) Seasonal (3-month) total of precipitation, seasonal average of Tmin, and seasonal average of Tmax, at 1-month lead-time (i.e. forecast for a (rolling) season beginning in 1-month's time).
- (3) Seasonal total of precipitation, seasonal average of Tmin, and seasonal average of Tmax, at 4-month lead-time (i.e. forecast for a (rolling) season beginning in 4-months' time).

The shorter target period of (1) is chosen because the forecasts for the first month ahead are the most skillful. The longer target periods of (2) and (3) are for covering the rest of the GCM forecast horizon with just two sets of evaluation. The evaluation is made at each of the 1348 grid cells (as in System 4) covering Australia. We evaluate both mean-corrected and BJP calibrated forecasts of SEAS5 and System 4. As raw forecasts are sometimes applied without any bias-correction, we additionally evaluate raw forecasts to confirm that it is not a good practice to

directly use raw forecasts.

To produce the mean-corrected forecasts, we apply a simple multiplicative mean-correction to precipitation, and an additive mean-correction to temperatures. Formulas for these simple mean-corrections are given in [Appendix B](#). The mean-corrections are made separately for individual grid cells and for individual months or seasons of the year, using a leave-one-year-out cross-validation setup. To produce calibrated forecasts for each of the target variables, we also establish and apply BJP models separately for individual grid cells and for individual months or seasons of the year, using a leave-one-year-out cross-validation setup.

We use the continuous ranked probability score (CRPS) as a measure of forecast error, corresponding to the difference between an ensemble forecast and its corresponding observation ([Hersbach, 2000](#)). A brief description of CRPS is given in [Appendix C](#). For CRPS to be low, ensemble forecasts need to be both accurate and have the right spread (not too narrow or too wide). We also evaluate CRPS for climatology ensemble forecasts (see [Appendix A](#) for more details about climatology ensemble forecasts). This second CRPS is used as a reference, so that percentage of reduction in average CRPS from the reference can be calculated to give a CRPS skill score (see [Appendix C](#)). A skill score of zero indicates that the forecasts are only as good as the reference forecasts. A negative (positive) skill score indicates that the forecasts are poorer (better) than the reference forecasts. Forecasts can reach a skill score of 100% when perfectly matching observations.

We use the PIT (probability integral transform) histograms to check ensemble forecast reliability, for example, whether forecasts are biased, too wide or too narrow in ensemble spread (as a measure of forecast uncertainty). A brief description of PIT histogram is given in [Appendix C](#). Further details on PIT calculation, including how to deal with forecasts with censor threshold values, can be found in [Wang et al. \(2009\)](#) and [Wang and Robertson \(2011\)](#). Further details on using PIT histograms as a diagnostic tool can be found in [Gneiting et al. \(2007\)](#). The PIT values for all grid cells and months or seasons are pooled together to construct the PIT histograms.

4. Results

4.1. Skill of monthly total (or average) forecasts with zero lead time

Precipitation. Results of CRPS skill score for mean-corrected and BJP calibrated forecasts based on SEAS5 are shown in [Fig. 1](#). The mean-corrected forecasts generally have positive CRPS skill score over most of Australia. The high skill area changes with season. There are small pockets of negative skill. These pockets are mostly removed after the BJP calibration, with climatology-like ensemble forecasts taking over, resulting in near zero skill score. The widespread positive skill of the mean-corrected forecasts is largely retained after the calibration, although there is a slight reduction in skill in some areas. The reduction in skill is likely caused by the reliability improvement due to the BJP calibration (see later Section 4.3). [Wilks \(2018\)](#) pointed out that the CRPS may reward forecasts that are narrower at the expense of reliability in ensemble spread. It is also possible that the training of a sophisticated calibration model may introduce error, lowering the skill score under cross-validation.

Focusing on the BJP calibrated forecasts, the average skill score is the highest for March and October. The high skill area is in the north and north-central Australia for March, and in south-central Australia in October. Initial conditions of atmosphere, land and ocean are all expected to influence the climate in the immediate month ahead.

Results of CRPS skill score difference between SEAS5 and System 4 are shown in [Fig. 2](#). SEAS5 is overall more skillful than System 4. The difference is more pronounced with the mean-corrected forecasts. The BJP calibration reduces the skill gap between the two forecasting systems.

Tmin and Tmax. Results of CRPS skill score for monthly average of

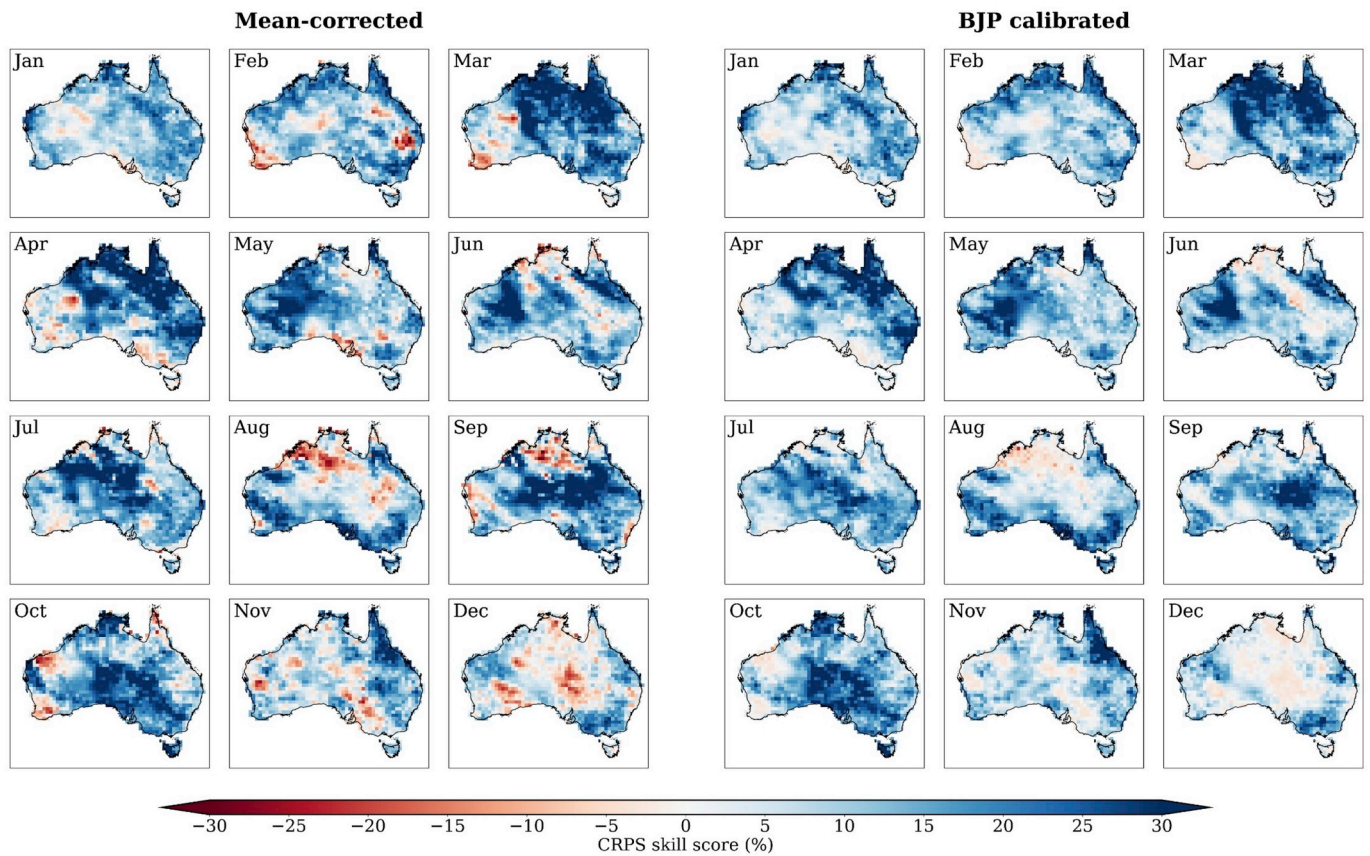


Fig. 1. CRPS skill score of SEAS5 forecasts of monthly precipitation with zero lead-time.

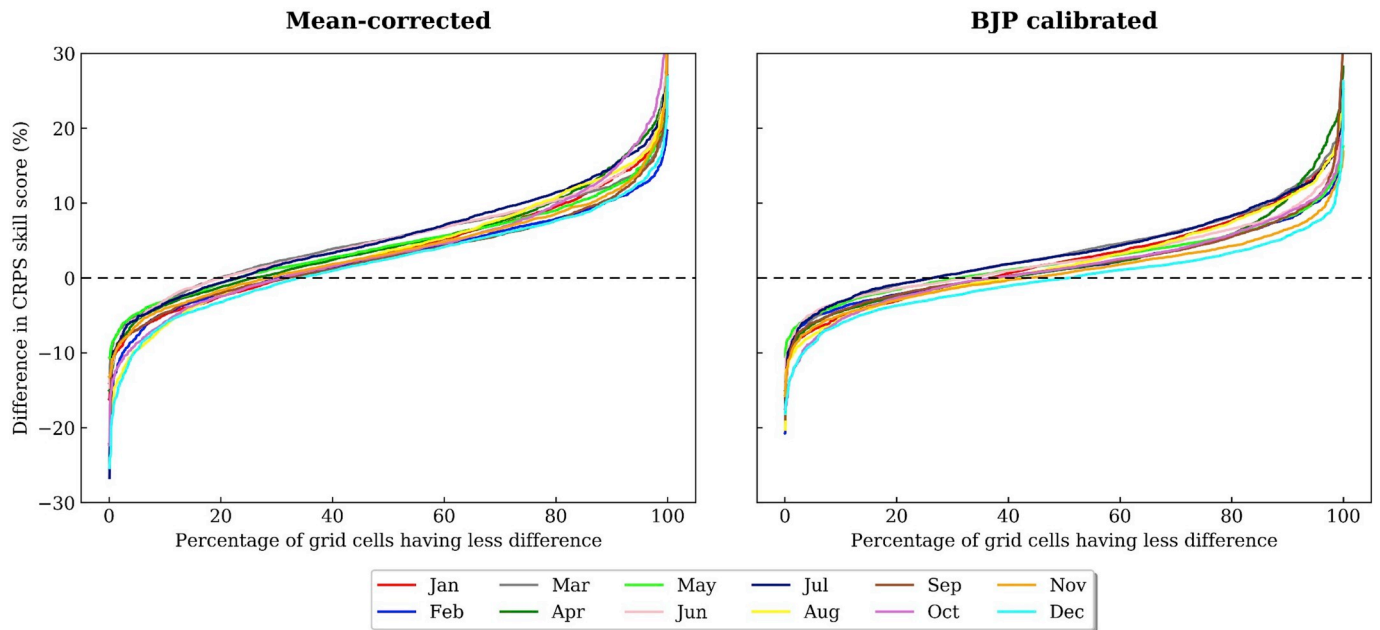


Fig. 2. Difference in CRPS skill score between SEAS5 and System 4 forecasts of monthly precipitation with zero lead-time.

Tmin forecasts based on SEAS5 are shown in Fig. 3. Both the mean-corrected forecasts and calibrated forecasts are more skillful overall than forecasts for precipitation. The mean-corrected forecasts have larger but fewer pockets of negative skill than forecasts for precipitation. The negative skill score can be below -30% . Again, the BJP calibration successfully removes the pockets of negative skill. The average skill

score of the BJP calibrated forecasts is the highest for March and October, as with precipitation. From the plots shown in Fig. 4 of the skill difference between the two forecasting systems, SEAS5 shows only a small overall improvement, with visible gain for some months but loss for some other months. After the BJP calibration, the difference between the two systems is overall unnoticeable, but variation with forecast

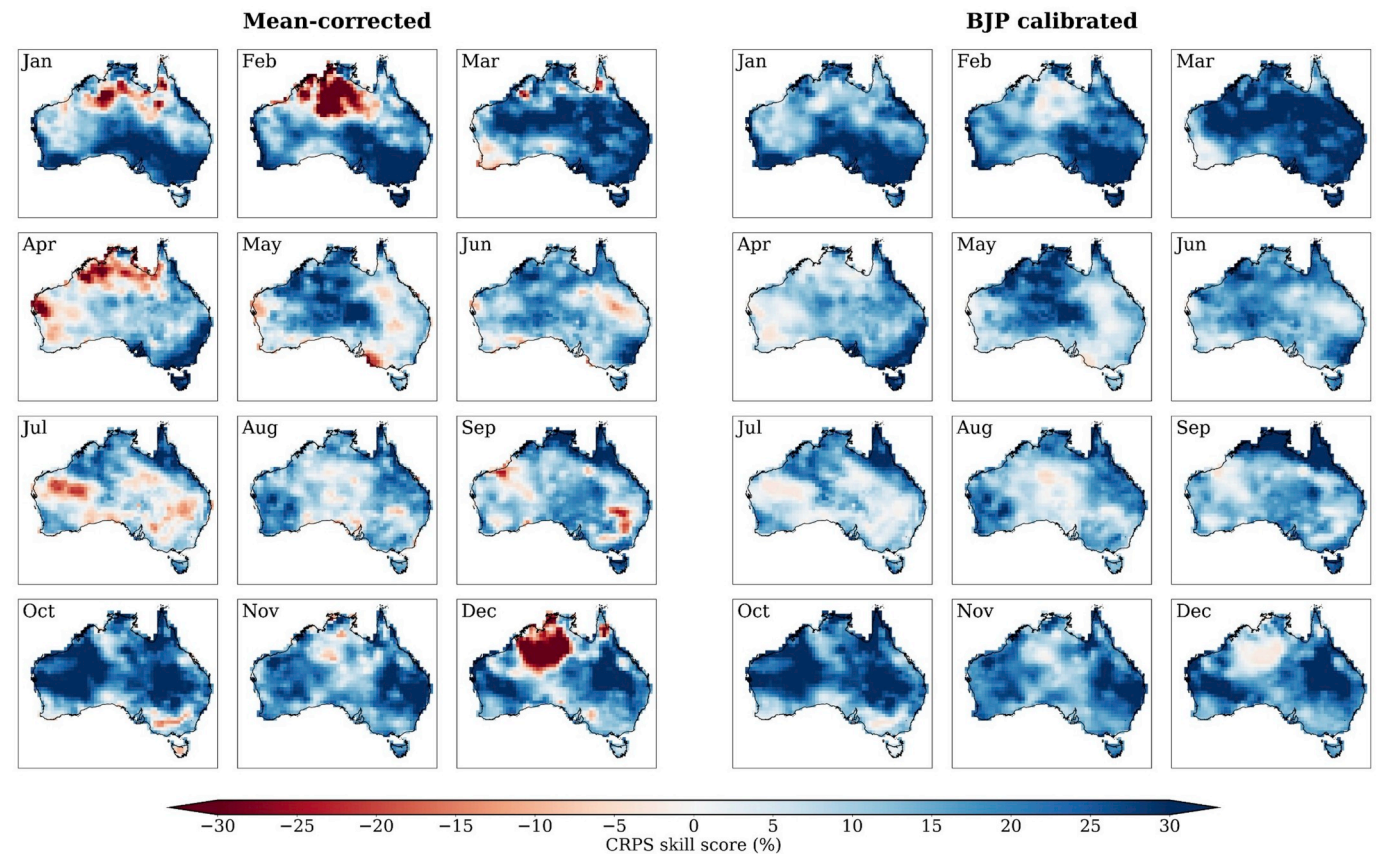


Fig. 3. CRPS skill score of SEAS5 forecasts of monthly average of Tmin with zero lead-time.

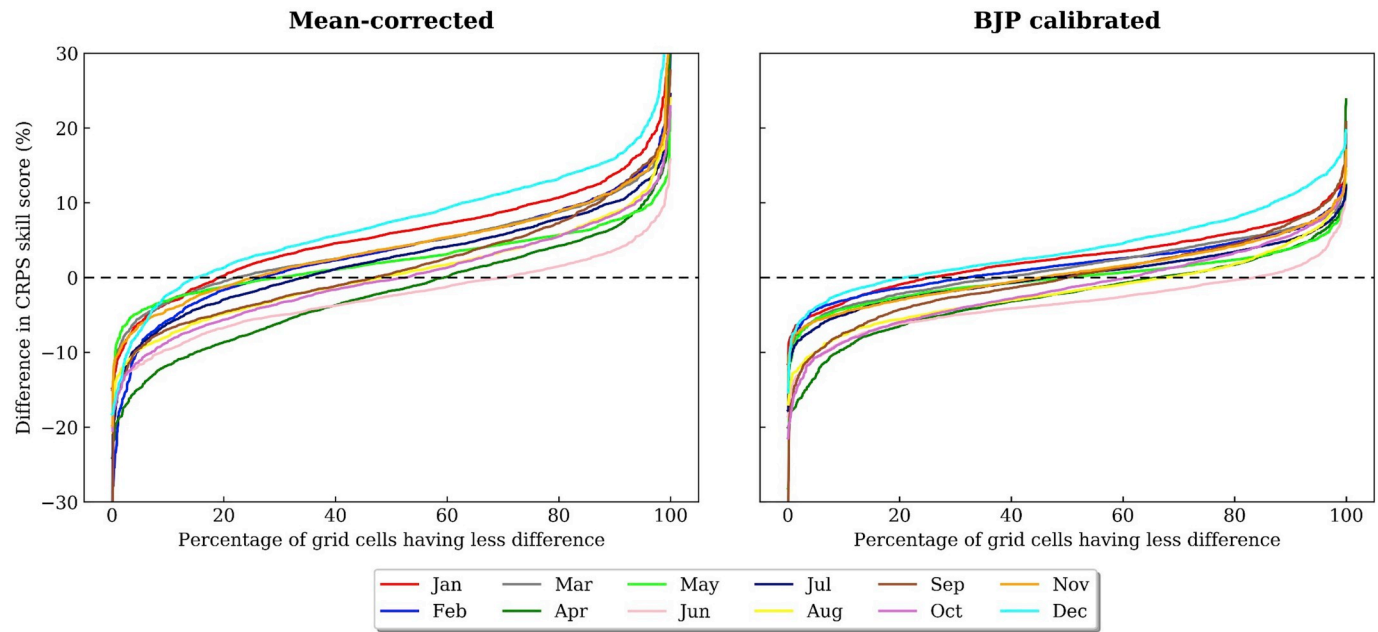


Fig. 4. Difference in CRPS skill score between SEAS5 and System 4 forecasts of monthly average of Tmin with zero lead-time.

month remains obvious (for example, gain for December and loss for June).

Results of evaluation of forecasts for Tmax are shown in Supplementary Material (Fig. S1 and Fig. S2). The CRPS skill score tends to be much higher than for Tmin forecasts. SEAS5 shows remarkable improvement over System 4. The average skill score of the BJP

calibrated forecasts is the highest for October, September, March and April.

4.2. Skill of seasonal total (or average) forecasts with 1-month lead time

Precipitation. Results of CRPS skill score for forecasts based on

SEAS5 are shown in Fig. 5. The skill score is low when compared with the monthly forecasts with zero lead-time (Fig. 1). Again, the BJP calibration removes the negative skill seen in the mean-corrected forecasts. The spatial pattern of areas of positive skill score of the BJP calibrated forecasts is remarkably consistent with the statistical evidence presented by Schepen et al. (2012, Figure 14) for using lagged climate indices to forecast seasonal rainfall over Australia. This consistency suggests that the influence of large global circulations on seasonal rainfall over Australia one-month ahead is (1) well modelled by SEAS5 and (2) responsible for nearly all the forecast skill.

From the skill score difference between the two forecasting systems shown in Fig. 6, SEAS5 shows improvement in mean-corrected forecasts over System 4 in most months, but the improvement is unnoticeable after the BJP calibration.

Tmin and Tmax. Results of CRPS skill score are shown in Fig. 7 for Tmin forecasts and in Fig. S3 for Tmax forecasts. Again, the BJP calibration removes the negative skill seen in the mean-corrected forecasts. The skill score is overall much lower than the monthly forecasts with zero lead-time (Fig. 3 and Fig. S1). Forecasts for Tmax are more skillful than for Tmin, and both are much better than for precipitation (Fig. 5).

From the skill score difference for Tmin between the two forecasting systems as shown in Fig. 8, SEAS5 shows overall improvement in mean-corrected forecasts over System 4, with visible gain for some seasons but loss for some other seasons. The improvement is less noticeable after the BJP calibration, but variation with forecast season remains obvious (for example, gain for Nov-Dec-Jan and loss for May-Jun-Jul). For Tmax, the improvement is more pronounced, and can be clearly seen even after the BJP calibration (Fig. S4).

4.3. Reliability of ensemble forecasts

PIT histograms are shown in Fig. 9 for precipitation and in Fig. 10 for

Tmin. The mean-corrected monthly forecasts of both SEAS5 and System 4 show a U-shape, indicating that the forecast ensemble spread is too narrow. The same can be said about the mean-corrected seasonal forecasts, but to a lesser extent (especially for precipitation). From results for Tmax shown in Supplementary Material (Fig. S5), the mean-corrected forecasts of both SEAS5 and System 4 are more reliable than Tmin in ensemble spread, especially the seasonal forecasts. In all cases, the BJP calibrated forecasts show a reasonably uniform shape, indicating that the calibrated forecast ensemble spread is reliable.

4.4. Other results

Results of CRPS skill score for mean-corrected and BJP calibrated seasonal forecasts based on SEAS5 at 4-month lead-time are provided in Supplementary Material (Figs. S6–S8). Compared with the seasonal forecasts at 1-month lead-time, the spatial extent and magnitude of positive skill score are reduced considerably. As before, among the three forecast variables, forecasts for Tmax are the most skillful, and forecasts for precipitation are the least skillful.

Raw SEAS5 forecasts for all variables and lead-times are found to be poor, often with large bias and highly negative skill score, especially for Tmin and Tmax (results not shown). The forecasts tend to be too high for Tmin, and too low for Tmax. We strongly discourage the direct use of raw forecasts.

5. Conclusions

For precipitation, Tmin and Tmax over the Australian continent, SEAS5 offers considerable skill for monthly forecasts with zero lead-time. The skill is the highest for Tmax and the lowest for precipitation. Compared with System 4, SEAS5 shows very large improvement to forecasts for Tmax, large improvement to precipitation, but little

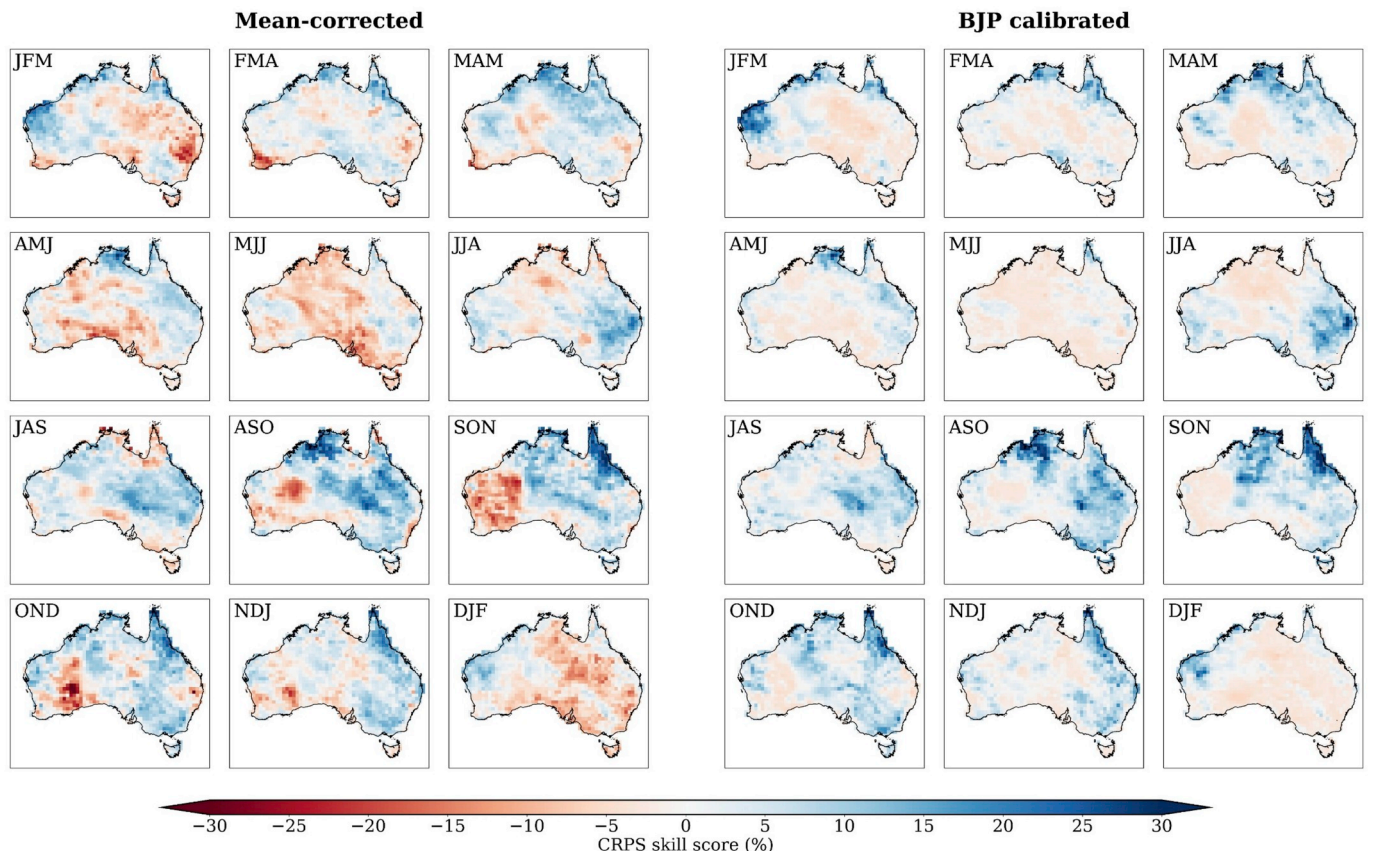


Fig. 5. CRPS skill score of SEAS5 forecasts of seasonal precipitation with 1-month lead-time.

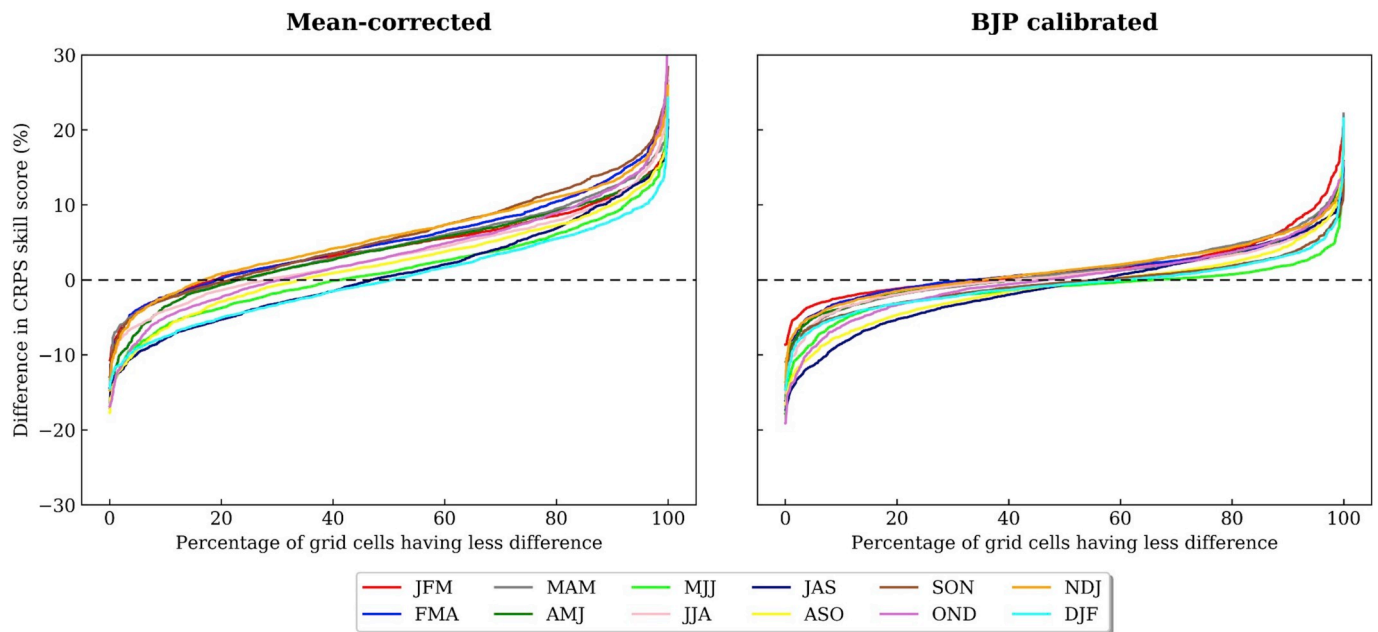


Fig. 6. Difference in CRPS skill score between SEAS5 and System 4 forecasts of seasonal precipitation with 1-month lead-time.

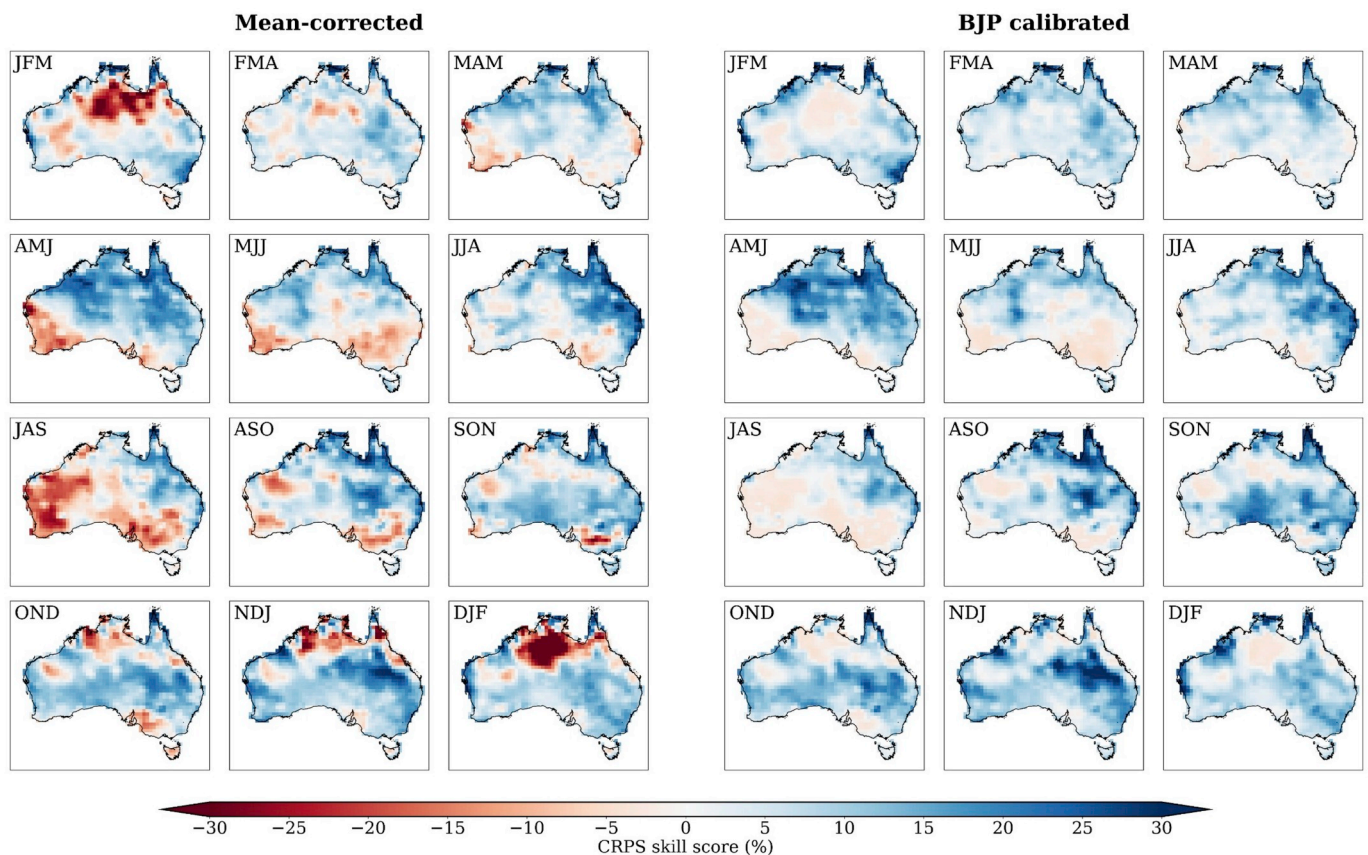


Fig. 7. CRPS skill score of SEAS5 forecasts of seasonal average of Tmin with 1-month lead-time.

improvement to Tmin. Overall, we suggest that the SEAS5 monthly forecasts at zero lead-time, that is forecasts at the start of a month for that month, can be valuable for applications in Australia. However, the skill varies across Australia, with high skill areas changing with time of the year. For precipitation and Tmin, the average skill score is the highest for March and October. For Tmax, the average skill score is the

highest for October, September, March and April.

SEAS5 forecast skill drops significantly beyond one month forecast horizon. The seasonal forecasts with 1-month lead-time are low in skill for precipitation, and very patchy for temperatures. This is despite some improvement over System 4, especially for Tmax. Overall, we suggest that SEAS5 temperature forecasts for beyond one month forecast

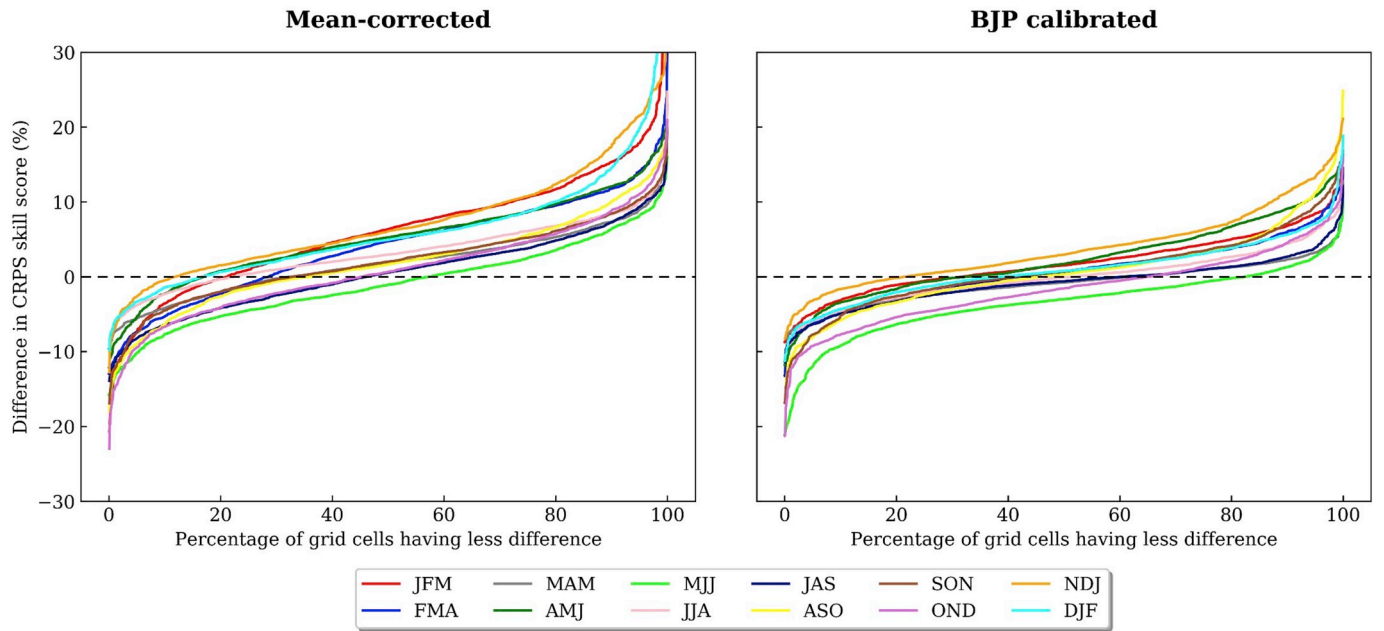


Fig. 8. Difference in CRPS skill score between SEAS5 and System 4 forecasts of seasonal average of Tmin with 1-month lead-time.

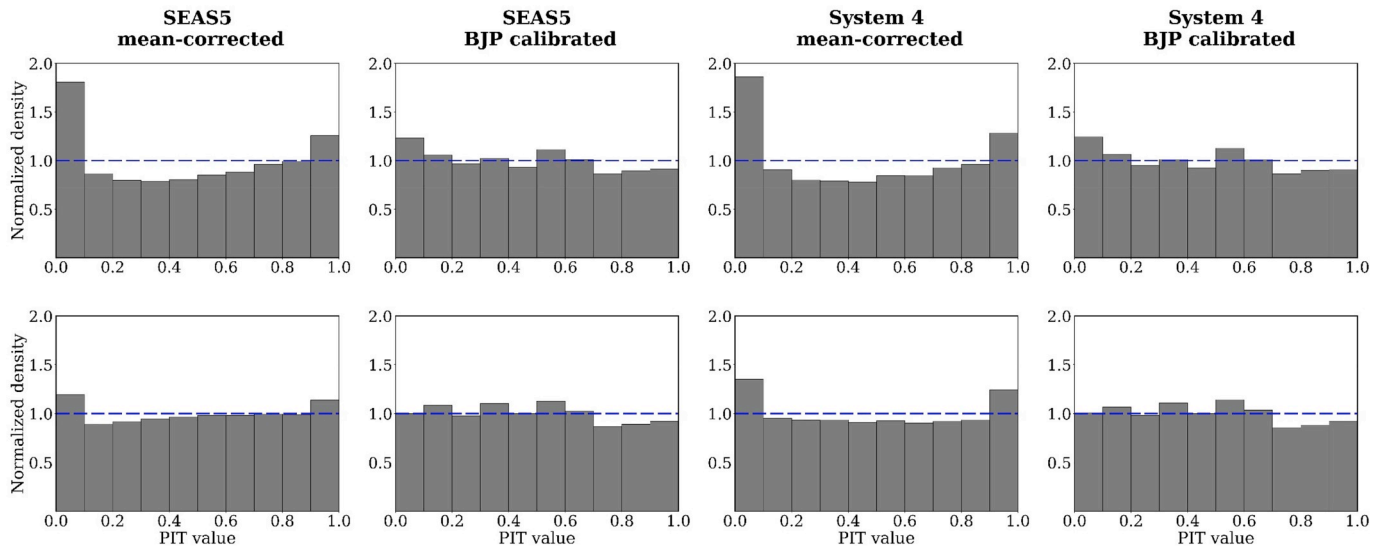


Fig. 9. PIT histograms of precipitation forecasts. Row 1 is for monthly forecasts with zero lead-time. Row 2 is for seasonal forecasts with 1-month lead-time.

horizon can be useful for only small parts of Australia, but precipitation forecasts offer little value.

Interestingly, the spatial pattern of areas of positive skill score of the BJP calibrated forecasts of seasonal precipitation total with 1-month lead-time is remarkably consistent with the statistical evidence presented by Schepen et al. (2012, Figure 14) for using lagged climate indices to forecast seasonal rainfall over Australia. This consistency suggests that the influence of large global circulations on seasonal rainfall over Australia one-month ahead is (1) well modelled by SEAS5 and (2) responsible for nearly all the forecast skill.

The mean-corrected SEAS5 forecasts are found to be unreliable in ensemble spread, generally too narrow. In other words, the forecasts are more confident than warranted. This is especially so for the monthly forecasts with zero lead-time. SEAS5 does not appear to show any improvement over System 4 in terms of reliability for Australian precipitation and temperatures. Another issue is that the mean-corrected SEAS5 forecasts are found to be less skillful than climatology

ensemble forecasts for some locations and times of the year.

The BJP calibration is found to be effective in making forecasts reliable and ensuring that forecasts are not worse than climatology ensemble forecasts, while retaining the positive skill of the GCM forecasts. We suggest that the calibrated ensemble forecasts are better forecasts for practical applications. By contrast, raw SEAS5 forecasts for all variables and lead-times are found to be poor, often with large bias and highly negative skill score. We strongly discourage the direct use of raw forecasts.

The new BJP algorithm introduced in this paper is much simplified mathematically and more straightforward to code up than the original BJP model. We find it highly efficient computationally in our post-processing of SEAS5 and System 4 forecasts for the very large number grid cells across the Australia continent and the leave-one-year-out cross-validation of 36 years of forecast data.

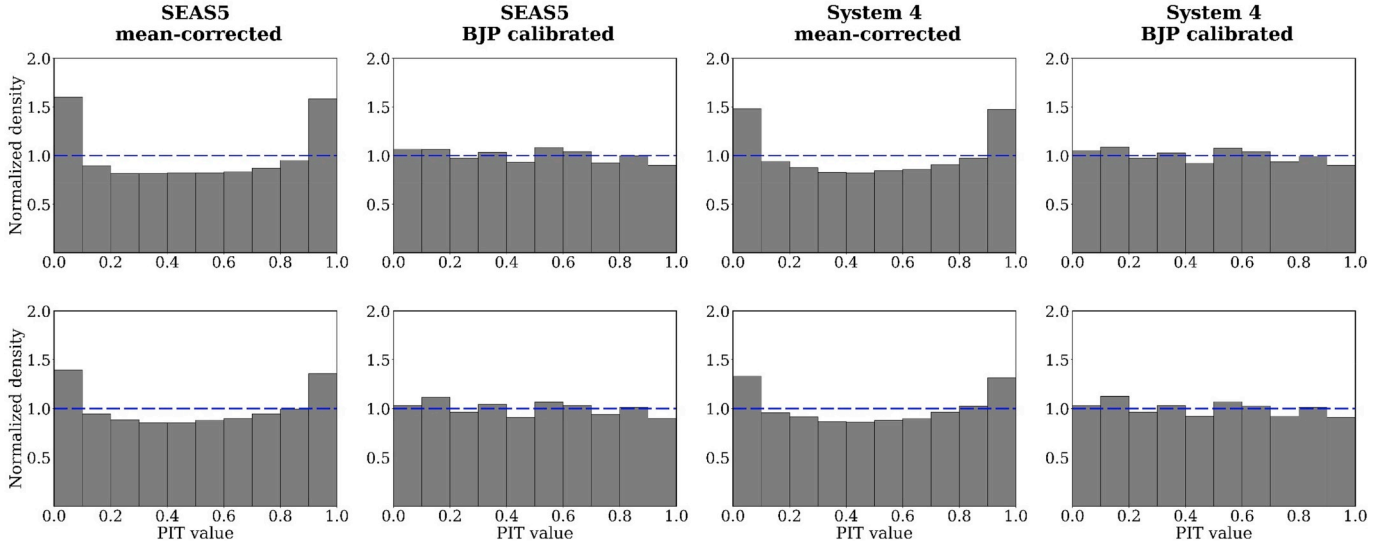


Fig. 10. PIT histograms of Tmin forecasts. Row 1 is for monthly forecasts with zero lead-time. Row 2 is for seasonal forecasts with 1-month lead-time.

Software availability

Pseudo computer code is given in [Appendix A](#). R and Python packages can be made available on request.

Acknowledgements

We thank the European Centre for Medium-Range Weather Forecasts

(ECMWF) for providing the SEAS5 and System 4 data. The data for SEAS5 can be obtained through the Copernicus Climate Data store (cds.climate.copernicus.eu). Florian Pappenberger acknowledges funding by the Copernicus program which supported the production of SEAS5 hindcasts. We thank the Bureau of Meteorology for AWAP data used in this study.

Appendix D. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envsoft.2019.104550>.

Appendix A

The Bayesian joint probability model – Parameter inference and predictive use via Gibbs sampling

In Section 3.1, we have presented the BJP model for a bivariate problem. The original BJP model (Wang et al., 2009; Wang and Robertson, 2011) was formulated as a multivariate problem, allowing for multiple predictors and multiple predictands. In this appendix, we also present the BJP model in a general multivariate formulation, of which the bivariate case is a special one.

A.1 The model

Consider d variables composed of both predictands and predictors, with n number of data records

$$\mathbf{z}(t) = [z_1(t) \ z_2(t) \ \dots \ z_d(t)]^T \quad (\text{A.1})$$

where $t = 1, 2, \dots, n$, $\mathbf{z}(t)$ is a column vector, and T is for transpose. We assume that the individual variables are normally-distributed or have been normalized through transformation, and $\mathbf{z}(t)$ is a sample drawn from a multivariate normal distribution

$$\mathbf{z}(t) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (\text{A.2})$$

where $\boldsymbol{\mu}$ is a $d \times 1$ mean vector, and $\boldsymbol{\Sigma}$ a $d \times d$ covariance matrix.

We assume that the variables may sometimes have missing values, or left censored values, for example $z_i(t) \leq z_i^c$.

A.2 Parameter inference

The posterior distribution of the model parameters is according to Bayes' theorem

$$p(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{D}) \propto p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) p(\mathbf{D} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \prod_{t=1}^n p(\mathbf{z}(t) | \boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (\text{A.3})$$

where $\mathbf{D} = \{\mathbf{z}(t), t = 1, 2, \dots, n\}$, $p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a prior distribution of the model parameters, and $p(\mathbf{D} | \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the likelihood. We apply the non-informative multivariate Jeffreys prior (Gelman et al., 2014)

$$p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \propto |\boldsymbol{\Sigma}|^{-(d+1)/2} \quad (\text{A.4})$$

By combining Eq. (A.2) – Eq. (A.4), we can derive following conditional distributions for the parameters (Gelman et al., 2014)

$$[\boldsymbol{\Sigma} | \cdot] = \text{Inv - Wishart}_{n-1}(\mathbf{S}) \quad (\text{A.5})$$

$$[\boldsymbol{\mu} | \cdot] = N(\bar{\mathbf{z}}, \boldsymbol{\Sigma} / n) \quad (\text{A.6})$$

where

$$\mathbf{S} = \sum_{t=1}^n (\mathbf{z}(t) - \bar{\mathbf{z}})(\mathbf{z}(t) - \bar{\mathbf{z}})^T \quad (\text{A.7})$$

$$\bar{\mathbf{z}} = \frac{1}{n} \sum_{t=1}^n \mathbf{z}(t) \quad (\text{A.8})$$

the symbol $|\cdot|$ means conditional on all other variables, and $\text{Inv - Wishart}_{n-1}$ is the Inverse-Wishart distribution with $n - 1$ degrees of freedom.

When one of variables $z_i(t)$ is missing a value, we can also derive its conditional distribution

$$[z_i(t) | \cdot] = N(\mu_i^*(t), \Sigma_{i,i}^*) \quad (\text{A.9})$$

where

$$\Sigma_{i,i}^* = \Sigma_{i,i} - \Sigma_{i,(i)} \left| \Sigma_{(i),(i)} \right|^{-1} \Sigma_{(i),i} \quad (\text{A.10})$$

$$\mu_i^*(t) = \mu_i + \Sigma_{i,(i)} \left| \Sigma_{(i),(i)} \right|^{-1} (\mathbf{z}_{(i)}(t) - \boldsymbol{\mu}_{(i)}) \quad (\text{A.11})$$

with (i) denoting all index entries in $\{1, 2, \dots, d\}$ except i .

The conditional distributions of Eq. (A.5) – Eq. (A.11) are used to set up Gibbs sampling to sequentially draw one sample from each of the conditional distributions, and repeat many cycles of such sampling. This yields a parameter chain, which when converged forms multiple parameter sets to numerically represent the joint posterior distribution of the parameters and the missing values.

Note that when $z_i(t)$ has a censored value instead of a missing value, we can also use the conditional distribution of Eq. (A.9) to sample a value. We just need to restrict the sampling to the range of $z_i(t) \leq z_i^c$.

Here we include pseudo code for implementing the Gibbs sampling for parameter inference.

Setting initial values

If the value of $z_i(t)$ is censored in the original data, set $z_i(t) = z_i^c$

If the value of $z_i(t)$ is missing in the original data, set $z_i(t) = \hat{z}_i$, where $\hat{z}_i$ is the average of non-missing

$z_i(t)$, $t = 1, \dots, n$

Gibbs sampling

Repeat

Compute $\mathbf{S}$ and $\bar{\mathbf{z}}$

Sample $\boldsymbol{\Sigma}$. Check if $\boldsymbol{\Sigma}$ is positive definite. If not, resample $\boldsymbol{\Sigma}$.

Sample $\boldsymbol{\mu}$.

Randomise the ordering $\{1, 2, \dots, d\}$ to $\text{ran_ord}()$, by sampling without replacement

Do $j=1, d$

$i=\text{ran_ord}(j)$

If any of the values of $z_i(t)$ $t = 1, \dots, n$, is missing or censored in the original data,

compute $\Sigma_{i,(i)} \left| \Sigma_{(i),(i)} \right|^{-1}$ and $\Sigma_{i,i}^*$

Do $t=1$ to n

If the value of $z_i(t)$ is missing in the original data, compute $\mu_i^*(t)$, sample

$[z_i(t) | \cdot] = N(\mu_i^*(t), \Sigma_{i,i}^*)$, and update $z_i(t)$ with the newly sampled value

If the value of $z_{i,d}$ is censored in the original data, compute $\mu_i^*(t)$, sample

$[z_i(t) | \cdot] = N(\mu_i^*(t), \Sigma_{i,i}^*)$, $z_i(t) \leq z_i^c$, and update $z_i(t)$ with the newly sampled value

End do

End do

End repeat

A.3 Prediction

Once samples of the posterior distribution of the parameters are available, we can apply the model for prediction. The vector $\mathbf{z}(t) = [z_1(t) \ z_2(t) \ \dots \ z_d(t)]^T$ consists of both predictor variables and predictand variables. In a prediction, we want to find the posterior distribution of the

predictands given information about the predictors. Sampling of this distribution can be conveniently done by treating the predictand variables as having missing values, and applying the Gibbs sampling step as outlined in Eq. (A.9) – Eq. (A.11). In fact, some of the predictors can also have missing or censored values and be included in the Gibbs sampling. For variables with censored values, we need to restrict the sampling to the range of $z_i(t) \leq z_i^c$ as discussed in Section A.2.

Here we include pseudo code for implementing the Gibbs sampling for prediction.

```

Setting initial values
  If the value of  $z_i(t)$  is censored in the original data,  $z_i(t) = z_i^c$ 
  If the value of  $z_i(t)$  is missing in the original data,  $z_i(t) = \mu_i$  (value of the first set of parameters)
Gibbs sampling
  Repeat sampling
    Use  $\mu$  and  $\Sigma$  already sampled
    Randomise the ordering  $\{1, 2, \dots, d\}$  to  $\text{ran\_ord}()$ , by sampling without replacement
    Do  $j=1, d$ 
       $i=\text{ran\_ord}(j)$ 
      If the value of  $z_i(t)$  is missing or censored in the original data, compute  $\mu_i^*(t)$  and  $\Sigma_{i,i}^*$ 
      If the value of  $z_i(t)$  is missing in the original data, sample  $[z_i(t) | \cdot] = N(\mu_i^*(t), \Sigma_{i,i}^*)$ , and
      update  $z_i(t)$  with the newly sampled value
      If the value of  $z_i(t)$  is censored in the original data, sample  $[z_i(t) | \cdot] = N(\mu_i^*(t), \Sigma_{i,i}^*)$ ,
       $z_i(t) \leq z_i^c$ , and update  $z_i(t)$  with the newly sampled value
    End do
  End repeat

```

A.4 Generating reference predictions

Sometimes it is useful to generate predictions using a naïve model to provide a reference for evaluating predictions from using a more advanced model. One such naïve model is simply based on the distribution of historical observations of the predictands.

Recall the full joint probability model of Eq. (A.2), we can reduce the model to a subset

$$\mathbf{z}^*(t) \sim N(\mu^*, \Sigma^*) \quad (\text{A.12})$$

where $\mathbf{z}^*(t)$ consists of only the predictand variables. Given the parameter samples already obtained through Gibbs sampling in Section A.2, we can generate samples of $\mathbf{z}^*(t)$ by sampling the distribution in Eq. (A.12). These samples can be used as ensemble reference predictions.

Here we include pseudo code for implementing the Gibbs sampling for generating reference predictions.

```

Gibbs sampling
  Sample only the predictand subset  $\mathbf{z}^*(t)$ 
  Repeat
    Use  $\mu^*$  and  $\Sigma^*$  already sampled, where  $\mu^*$  and  $\Sigma^*$  are the subset of  $\mu$  and  $\Sigma$  corresponding
    to  $\mathbf{z}^*(t)$ 
    Directly sample  $[\mathbf{z}^*(t)] = N(\mu^*, \Sigma^*)$ , rather than by Gibbs sampling
  End repeat

```

Appendix B

Simple bias-corrections

For precipitation, we use a multiplicative bias-correction

$$y_2(t^*) = \frac{\sum_{t=1}^n y_2(t)}{\sum_{t=1}^n y_1(t)} y_1(t^*) \quad (\text{B.1})$$

For temperature, we use an additive bias-correction

$$y_2(t^*) = y_1(t^*) + \left[\frac{1}{n} \sum_{t=1}^n y_2(t) - \frac{1}{n} \sum_{t=1}^n y_1(t) \right] \quad (\text{B.2})$$

where $y_2(t)$ denotes observed data, $y_1(t)$ denotes raw ensemble forecast mean, n is the number of events used in model fitting, $y_1(t^*)$ is a new forecast ensemble member, and $y_2(t^*)$ is a bias-corrected forecast ensemble member. The bias-correction is applied to all members.

Appendix C

CRPS skill score and PIT histogram

The continuous ranked probability score (CRPS; Hersbach, 2000) is adopted to evaluate the forecast skill. The average CRPS for time steps $t = 1, 2, \dots, n$ is formulated as:

$$CRPS = \frac{1}{n} \sum_{t=1}^n \int [F(t, x) - H(x - y(t))]^2 dx \quad (C.1)$$

where $F(t, x)$ is the cumulative distribution function (CDF) of an ensemble forecast at time t for variable x , $y(t)$ is the observed value, and H is the Heaviside step function which equals 0 if $x < y(t)$ and equals 1 otherwise. CRPS is also calculated for a set of reference forecasts (see Section 3.2). A skill score is then computed as:

$$CRPS \text{ skill score} = \frac{CRPS_{ref} - CRPS}{CRPS_{ref}} \times 100 (\%) \quad (C.2)$$

Reliability of the ensemble forecasts is evaluated by calculating the forecast probability integral transform (PIT) values and plotting the probability integral transform (PIT) histogram (Gneiting et al., 2007). For each pair of forecast and observation, the PIT value is calculated by,

$$PIT(t) = \begin{cases} F(t, y(t)) & y(t) > 0 \\ a \times F(t, 0) & y(t) = 0 \end{cases}$$

where a is randomly sampled from the uniform distribution $[0, 1]$ (Wang and Robertson, 2011). Theoretically, if the probabilistic forecasts are perfectly reliable, the collection of PIT values should follow a standard uniform distribution. In this study, the PIT histogram is generated by pooling the pairwise forecasts and observations for all grids and all months or seasons. The reliability can be examined by inspecting the uniformity of the PIT histogram. The more uniform the histogram is, the more reliable the forecasts are.

References

- Bennett, J.C., Wang, Q.J., Li, M., Robertson, D.E., Schepen, A., 2016. Reliable long-range ensemble 5 streamflow forecasts: combining calibrated climate forecasts with a conceptual runoff model and a staged error model. *Water Resour. Res.* 52, 8238–8259. <https://doi.org/10.1002/2016WR019193>.
- Bennett, J.C., Wang, Q.J., Robertson, D.E., Schepen, A., Li, M., Michael, K., 2017. Assessment of an ensemble seasonal streamflow forecasting system for Australia. *Hydrol. Earth Syst. Sci.* 21, 6007–6030. <https://doi.org/10.5194/hess-21-6007-2017>.
- Emerton, R., Zsoter, E., Arnal, L., Cloke, H., Muraro, D., Prudhomme, C., Stephens, E.M., Salamon, P., Pappenberger, F., 2018. Developing a global operational seasonal hydro-meteorological forecasting system: GloFAS-Seasonal v1.0. *Geosci. Model Dev.* 11, 3327–3346. <https://doi.org/10.5194/gmd-11-3327-2018>, 2018.
- Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B., 2014. *Bayesian Data Analysis*, third ed. CRC press, Boca Raton.
- Gneiting, T., Raftery, A.E., Westveld III, A.H., Goldman, T., 2005. Calibrated probabilistic forecasting using ensemble model output statistics and minimum crps estimation. *Mon. Weather Rev.* 133, 1098–1118. <https://doi.org/10.1175/MWR2904.1>.
- Gneiting, T., Balabdaoui, F., Raftery, A.E., 2007. Probabilistic forecasts, calibration and sharpness. *J. R. Stat. Ser. Soc. B Stat. Methodol.* 69, 243–268. <https://doi.org/10.1111/j.1467-9868.2007.00587.x>.
- Hawthorne, S., Wang, Q.J., Schepen, A., Roberson, D.E., 2013. Effective use of general circulation model outputs for forecasting monthly rainfalls to long lead times. *Water Resour. Res.* 49, 5427–5436. <https://doi.org/10.1002/wrcr.20453>.
- Hersbach, H., 2000. Decomposition of the continuous ranked probability score for ensemble prediction systems. *Weather Forecast.* 15, 559–570. [https://doi.org/10.1175/1520-0434\(2000\)015<0559:DOTCRP>2.0.CO;2](https://doi.org/10.1175/1520-0434(2000)015<0559:DOTCRP>2.0.CO;2).
- Hudson, D., Alves, O., Hendon, H.H., Lim, E., Liu, G., Luo, J.-J., MacLachlan, C., Marshall, A.G., Shi, L., Wang, G., Wedd, R., Young, G., Zhao, M., Zhou, X., 2017. ACCESS-S1: the new Bureau of Meteorology multi-week to seasonal prediction system. *J. South. Hemisphere Earth Syst. Sci.* 67 (3), 132–159. <https://doi.org/10.2449/3.6703.001>.
- Johnson, S.J., Stockdale, T.N., Ferranti, L., Balmaseda, M.A., Molteni, F., Magnusson, L., Tietsche, S., Decremier, D., Weisheimer, A., Balsamo, G., Keeley, S., Mogenssen, K., Zuo, H., Monge-Sanz, B., 2018. SEASS: the new ECMWF seasonal forecast system. *Geosci. Model Dev. Discuss.* <https://doi.org/10.5194/gmd-2018-228> submitted for publication).
- Jones, D.A., Wang, W., Fawcett, R., 2009. High-quality spatial climate data-sets for Australia. *Aust. Meteorol. Oceanogr.* 58, 233–248.
- Kim, H.-M., Webster, P.J., Curry, J.A., 2012. Seasonal prediction skill of ECMWF system 4 and NCEP CFSv2 retrospective forecast for the northern Hemisphere winter. *Clim. Dyn.* 39, 2957–2973. <https://doi.org/10.1007/s00382-012-1364-6>.
- Molteni, F., Stockdale, T., Balmaseda, M., Balsamo, G., Buizza, R., Ferranti, L., Magnusson, L., Mogenssen, K., Palmer, T., Vitart, F., 2011. The new ECMWF seasonal forecast system (System 4), p. 656. *ECMWF Technical Memorandum*.
- Narapuseetty, B., Collins, D., Murtugudde, R., Gottschalk, J., Peters-Lidard, C., 2018. Bias correction to improve the skill of summer precipitation forecasts over contiguous United States by the North American Multi-Model Ensemble system. *Atmos. Sci. Lett.* 19, e818. <https://doi.org/10.1002/asl.818>.
- Peng, Z., Wang, Q., Bennett, J.C., Schepen, A., Pappenberger, F., Pokhrel, P., Wang, Z., 2014. Statistical calibration and bridging of ECMWF System4 outputs for forecasting seasonal precipitation over China. *J. Geophys. Res.: Atmospheres* 119, 7116–7135. <https://doi.org/10.1002/2013JD021162>.
- Schepen, A., Wang, Q., 2014. Ensemble forecasts of monthly catchment rainfall out to long lead times by post-processing coupled general circulation model output. *J. Hydrol.* 519, 2920–2931. <https://doi.org/10.1016/j.jhydrol.2014.03.017>.
- Schepen, A., Wang, Q.J., Robertson, D., 2012. Evidence for using lagged climate indices to forecast Australian seasonal rainfall. *J. Clim.* 25 (4), 1230–1246. <https://doi.org/10.1175/JCLI-D-11-00156.1>.
- Schepen, A., Wang, Q.J., Robertson, D.E., 2014. Seasonal forecasts of Australian rainfall through calibration and bridging of coupled GCM outputs. *Mon. Weather Rev.* 142 (5), 1758–1770. <https://doi.org/10.1175/MWR-D-13-00248.1>.
- Schepen, A., Wang, Q.J., Everingham, Y., 2016. Calibration, bridging, and merging to improve GCM seasonal temperature forecasts in Australia. *Mon. Weather Rev.* 144 (6), 2421–2441. <https://doi.org/10.1175/MWR-D-15-0384.1>.
- Schepen, A., Zhao, T., Wang, Q.J., Robertson, D.E., 2018. A Bayesian modelling method for post-processing daily sub-seasonal to seasonal rainfall forecasts from global climate models and evaluation for 12 Australian catchments. *Earth Syst. Sci.* 22, 1615–1628. <https://doi.org/10.5194/hess-22-1615-2018>.
- Shukla, S., Lettenmaier, D.P., 2013. Multi-RCM ensemble downscaling of NCEP CFS winter season forecasts: implications for seasonal hydrologic forecast skill. *J. Geophys. Res. Atmos.* 118 <https://doi.org/10.1002/jgrd.50628>, 10,770–10,790.
- Strazzo, S., Collins, D.C., Schepen, A., Wang, Q.J., Becker, E., Jia, L., 2018. Application of a hybrid statistical–dynamical system to seasonal prediction of north American temperature and precipitation. *Mon. Weather Rev.* 147, 607–625. <https://doi.org/10.1175/MWR-D-18-0156.1>.
- Troccoli, A., 2010. Seasonal climate forecasting. *Meteorol. Appl.* 17 (3), 251–268. <https://doi.org/10.1002/met.184>.
- Unger, D., Van den Dool, H., O'Lenic, E., Collins, D., 2009. Ensemble regression. *Mon. Weather Rev.* 137, 2365–2379. <https://doi.org/10.1175/2008MWR2605.1>.
- Wang, Q.J., Robertson, D.E., Chiew, F.H.S., 2009. A Bayesian joint probability modeling approach for seasonal forecasting of streamflows at multiple sites. *Water Resour. Res.* 45 <https://doi.org/10.1029/2008WR007355>. W05407.
- Wang, Q., Robertson, D., 2011. Multisite probabilistic forecasting of seasonal flows for streams with zero value occurrences. *Water Resour. Res.* 47 <https://doi.org/10.1029/2010WR009333>. W02546.

- Wang, Q.J., Shrestha, D.L., Robertson, D.E., Pokhrel, P., 2012. A log-sinh transformation for data normalization and variance stabilization. *Water Resour. Res.* 48 <https://doi.org/10.1029/2011WR010973>. W05514.
- Weisheimer, A., Palmer, T.N., 2014. On the reliability of seasonal climate forecasts. *J. R. Soc. Interface* 11 (96). <https://doi.org/10.1098/rsif.2013.1162>, 20131162.
- Wilks, D.S., 2018. Enforcing calibration in ensemble postprocessing. *Q. J. R. Meteorol. Soc.* 710, 76–84. <https://doi.org/10.1002/qj.3185>.
- Wilks, D.S., Hamill, T.M., 2007. Comparison of ensemble-MOS methods using GFS reforecasts. *Mon. Weather Rev.* 135, 2379–2390. <https://doi.org/10.1175/MWR3402.1>.
- Yeo, I.K., Johnson, R.A., 2000. A new family of power transformations to improve normality or symmetry. *Biometrika* 87 (4), 954–959.
- Zhang, W., Villarini, G., Slater, L., Vecchi, G.A., Bradley, A.A., 2017. Improved ENSO forecasting using bayesian updating and the north American multimodel ensemble (NMME). *J. Clim.* 30, 9007–9025. <https://doi.org/10.1175/JCLI-D-17-0073.1>.
- Zhao, T., Bennett, J., Wang, Q.J., Schepen, A., Wood, A., Robertson, D., Ramos, M.-H., 2017. How suitable is quantile mapping for post-processing GCM precipitation forecasts? *J. Clim.* 30, 3185–3196. <https://doi.org/10.1175/JCLI-D-16-0652.1>.
- Zhao, T., Wang, Q.J., Schepen, A., Griffith, M., 2019. Ensemble forecasting of monthly and seasonal reference crop evapotranspiration based on global climate model outputs. *Agric. For. Meteorol.* 264, 114–124. <https://doi.org/10.1016/j.agrformet.2018.10.001>.
- Zuo, H., Balmaseda, M.A., Mogensen, K., Tietsche, S., 2018. OCEAN5: the ECMWF ocean reanalysis system ORAS5 and its real-time analysis component, p. 823. ECMWF Technical Memorandum.